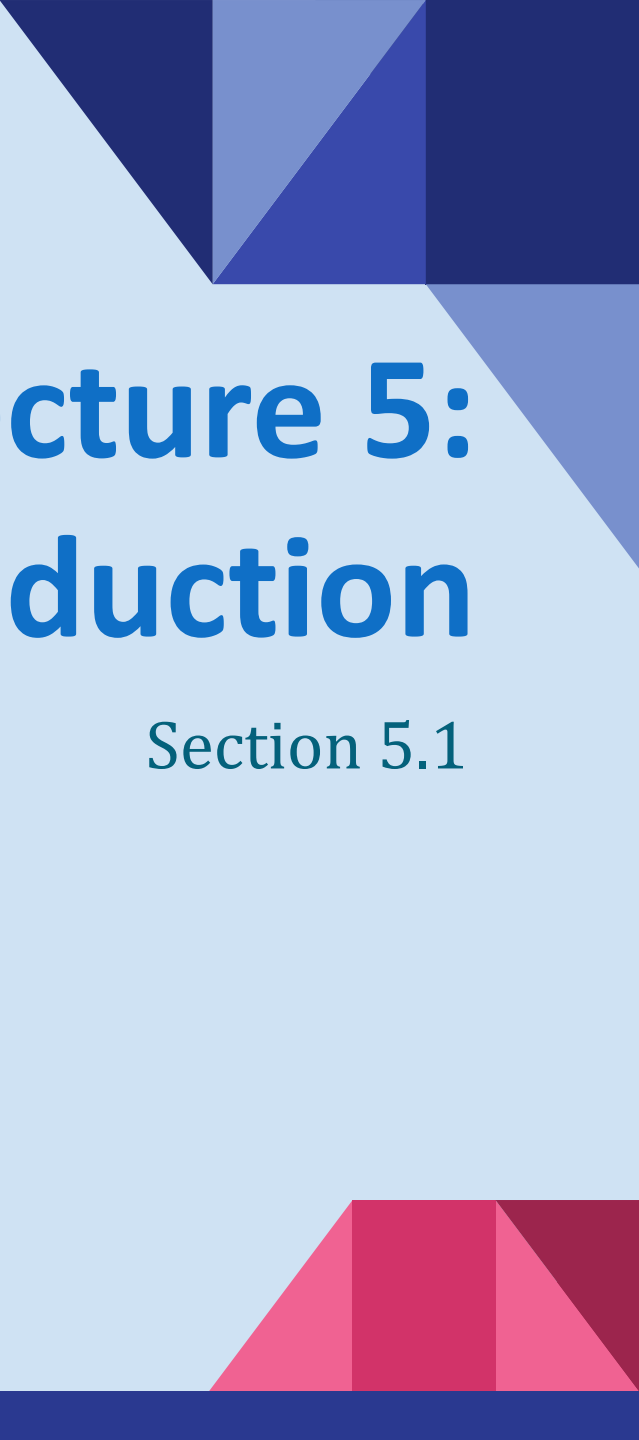


The slide features a light blue background. In the top right corner, there is a cluster of overlapping triangles in various shades of blue and purple. In the bottom right corner, there is a cluster of overlapping triangles in shades of pink and red. A solid dark blue horizontal bar runs across the bottom of the slide.

Lecture 5: Mathematical Induction

Section 5.1

Section Summary

- Mathematical Induction
- Examples of Proof by Mathematical Induction
- Mistaken Proofs by Mathematical Induction
- Guidelines for Proofs by Mathematical Induction

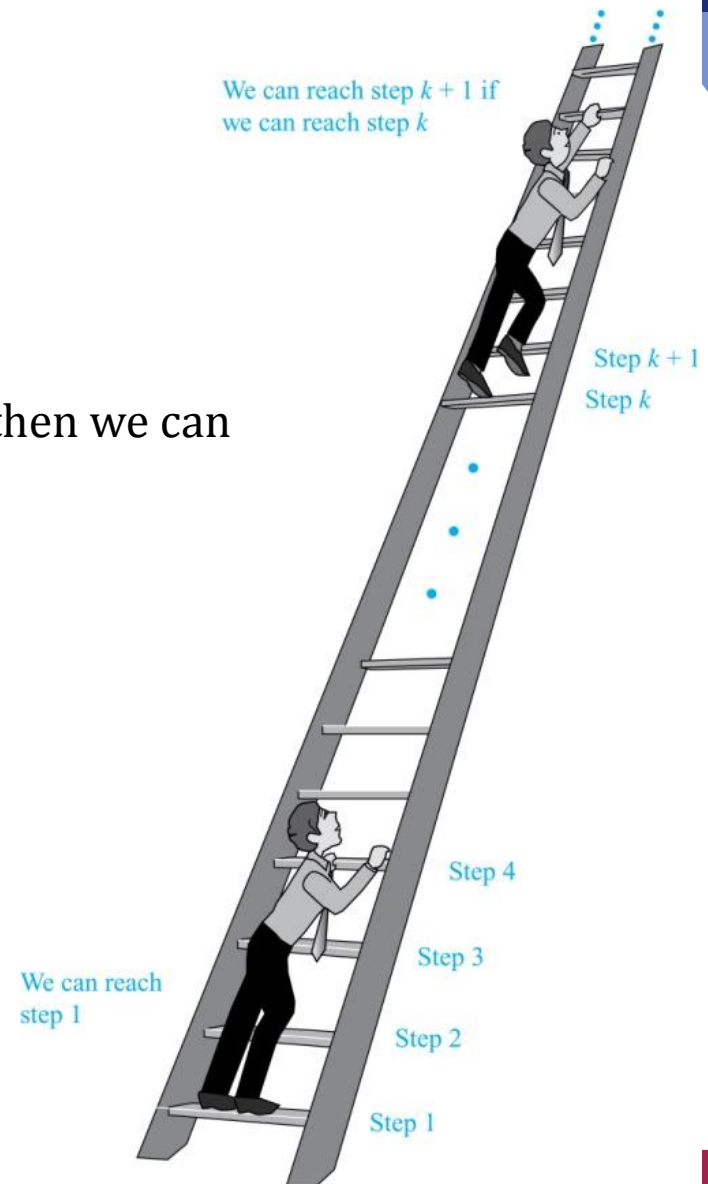
Climbing an Infinite Ladder

Suppose we have an infinite ladder:

1. We can reach the first rung of the ladder.
2. If we can reach a particular rung of the ladder, then we can reach the next rung.

From (1), we can reach the first rung. Then by applying (2), we can reach the second rung. Applying (2) again, the third rung. And so on. We can apply (2) any number of times to reach any particular rung, no matter how high up.

This example motivates proof by mathematical induction.



Principle of Mathematical Induction

Principle of Mathematical Induction: To prove that $P(n)$ is true for all positive integers n , we complete these steps:

- *Basis Step:* Show that $P(1)$ is true.
- *Inductive Step:* Show that $P(k) \rightarrow P(k + 1)$ is true for all positive integers k .

To complete the inductive step, assuming the *inductive hypothesis* that $P(k)$ holds for an arbitrary integer k , show that must $P(k + 1)$ be true.

Climbing an Infinite Ladder Example :

- BASIS STEP: By (1), we can reach rung 1.
- INDUCTIVE STEP: Assume the inductive hypothesis that we can reach rung k . Then by (2), we can reach rung $k + 1$.

Hence, $P(k) \rightarrow P(k + 1)$ is true for all positive integers k . We can reach every rung on the ladder. ◀

Important Points About Using Mathematical Induction

- Mathematical induction can be expressed as :

$$(P(1) \wedge \forall k (P(k) \rightarrow P(k + 1))) \rightarrow \forall n P(n),$$

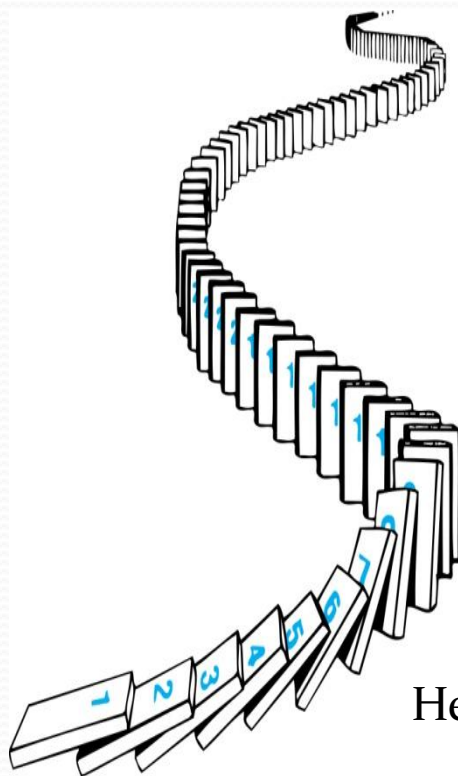
where the domain is the set of positive integers.

- In a proof by mathematical induction, we **don't prove** that $P(k)$ is true for all positive integers! We show that if we assume that $P(k)$ is true, then $P(k + 1)$ must also be true.
- Proofs by mathematical induction do not always start at the integer 1. In such a case, the basis step begins at a starting point b where b is an integer. We will see examples of this soon.

How Mathematical Induction Works

Consider an infinite sequence of dominoes, labeled $1, 2, 3, \dots$, where each domino is standing.

Let $P(n)$ be the proposition that the n th domino is knocked over.



We know that the first domino is knocked down, i.e., $P(1)$ is true.

We also know that if whenever the k^{th} domino is knocked over, it knocks over the $(k + 1)^{\text{st}}$ domino, i.e, $P(k) \rightarrow P(k + 1)$ is true for all positive integers k .

Hence, all dominos are knocked over.

$P(n)$ is true for all positive integers n .

Proving a Summation Formula by Mathematical Induction

Example : Show that if n is a positive integer, then:

$$\mathbf{P(n):} \quad 1 + 2 + \dots + n = \frac{n(n + 1)}{2}.$$

Solution :

- BASIS STEP: $P(1)$ is true since $1 = 1(1 + 1)/2$.
- INDUCTIVE STEP: Assume $P(k)$ holds.

Then, the inductive hypothesis is: $1 + 2 + \dots + k = \frac{k(k + 1)}{2}$

Under this assumption,

$$\begin{aligned} 1 + 2 + \dots + k + (k + 1) &= \frac{k(k + 1)}{2} + (k + 1) \\ &= \frac{k(k + 1) + 2(k + 1)}{2} \\ &= \frac{(k + 1)(k + 2)}{2} \end{aligned}$$



Conjecturing and Proving Correct a Summation Formula

Example : Conjecture and prove correct a formula for the sum of the first n positive odd integers. Then prove your conjecture.

Solution : We have: $1 = 1$, $1 + 3 = 4$, $1 + 3 + 5 = 9$, $1 + 3 + 5 + 7 = 16$,
 $1 + 3 + 5 + 7 + 9 = 25$.

- We can conjecture that the sum of the first n positive odd integers is n^2 ,
 $1 + 3 + 5 + \dots + (2n - 1) = n^2$.

- We prove that the conjecture is correct with mathematical induction.
- BASIS STEP: $P(1)$ is true since $1^2 = 1$.
- INDUCTIVE STEP: We have to show that, for every positive integer k ,
 $P(k) \rightarrow P(k + 1)$.

Assume the inductive hypothesis $P(k)$ is True and then show that $P(k + 1)$ holds as well.

(continued $\rightarrow$)

Conjecturing and Proving Correct a Summation Formula

Inductive Hypothesis : $P(k) \equiv "1 + 3 + 5 + \dots + (2k - 1) = k^2"$

- So, assuming $P(k)$, it follows that:

$$\begin{aligned} & 1 + 3 + 5 + \dots + (2k - 1) + (2k + 1) \\ &= [1 + 3 + 5 + \dots + (2k - 1)] + (2k + 1) \\ &= k^2 + (2k + 1) \text{ [by the inductive hypothesis]} \\ &= k^2 + 2k + 1 = (k + 1)^2 \end{aligned}$$

- Hence, we have shown that $P(k + 1)$ follows from $P(k)$.
Therefore the sum of the first n positive odd integers is n^2 .



More Summation Proof by Induction

Example :

Show that: Sum of a finite number of terms in a geometric progression with initial term a and common ratio r is

$$\sum_{j=0}^n ar^j = a + ar + ar^2 + \cdots + ar^n = \frac{ar^{n+1} - a}{r - 1} \quad \text{when } r \neq 1,$$

where n is a nonnegative integer.

Solution :

- BASIS STEP: $P(0)$ is true since $\frac{ar^{0+1} - a}{r - 1} = \frac{ar - a}{r - 1} = \frac{a(r - 1)}{r - 1} = a.$
- INDUCTIVE STEP: Assume true for $P(k)$. So the inductive hypothesis would be:

$$a + ar + ar^2 + \cdots + ar^k = \frac{ar^{k+1} - a}{r - 1}.$$

Under such assumption, we add ar^{k+1} to both sides of the equality above.

More Summation Proof by Induction (contd.)

Rewriting the R.H.S of the equation gives

$$\begin{aligned}\frac{ar^{k+1} - a}{r - 1} + ar^{k+1} &= \frac{ar^{k+1} - a}{r - 1} + \frac{ar^{k+2} - ar^{k+1}}{r - 1} \\ &= \frac{ar^{k+2} - a}{r - 1}.\end{aligned}$$

which implies- $a + ar + ar^2 + \dots + ar^k + ar^{k+1} = \frac{ar^{k+2} - a}{r - 1}.$

Conclusion: If the inductive hypothesis $P(k)$ is true, then $P(k+1)$ must be true. Proof by induction done!



Proving Divisibility Results

Example : Use mathematical induction to prove that $n^3 - n$ is divisible by 3, for every positive integer n .

Solution : Let $P(n)$ be the proposition that $n^3 - n$ is divisible by 3.

- BASIS STEP: $P(1)$ is true since $1^3 - 1 = 0$, which is divisible by 3.
- INDUCTIVE STEP: Assume $P(k)$ holds, i.e., $k^3 - k$ is divisible by 3, for an arbitrary positive integer k . To show that $P(k + 1)$ follows:

$$\begin{aligned}(k + 1)^3 - (k + 1) &= (k^3 + 3k^2 + 3k + 1) - (k + 1) \\ &= (k^3 - k) + 3(k^2 + k)\end{aligned}$$

By the inductive hypothesis, the first term $(k^3 - k)$ is divisible by 3 and the second term is divisible by 3 since it is an integer multiplied by 3. So by part (i) of Theorem 1 in Section 4.1, $(k + 1)^3 - (k + 1)$ is divisible by 3.

Therefore, $n^3 - n$ is divisible by 3, for every integer positive integer n . ◀

Proving More Divisibility

Example : Use mathematical induction to prove that $7^{n+2} + 8^{2n+1}$ is divisible by 57 for every nonnegative integer n .

Solution : Let $P(n)$ denote the proposition: “ $7^{n+2} + 8^{2n+1}$ is divisible by 57.”

- BASIS STEP: **$P(0)$** is true since $7^{0+2} + 8^{2 \cdot 0 + 1} = 7^2 + 8^1 = 57$ is divisible by 57.
- INDUCTIVE STEP: Assume $P(k)$ holds, that is, we assume that $7^{k+2} + 8^{2k+1}$ is divisible by 57. So we can write: $7^{k+2} + 8^{2k+1} = 57x$, where x is an integer.

Then, we must show that $P(k+1)$, the statement that

“ $7^{(k+1)+2} + 8^{2(k+1)+1}$ is divisible by 57”, is also true. We find that,

$$7^{(k+1)+2} + 8^{2(k+1)+1} = 7^{k+3} + 8^{2k+3} = 7 \cdot 7^{k+2} + 8^2 \cdot 8^{2k+1} = 7 \cdot 7^{k+2} + 64 \cdot 8^{2k+1}$$

$$= 7(7^{k+2} + 8^{2k+1}) + 57 \cdot 8^{2k+1} = 7(57x) + 57 \cdot 8^{2k+1} \quad \blacktriangleleft \quad = 57(7x + 8^{2k+1}).$$

Therefore, $7^{(k+1)+2} + 8^{2(k+1)+1}$ is divisible by 57.

An Incorrect “Proof” by Mathematical Induction (Cont.)

Example : Let $P(n)$ be the statement that “Every set of n lines in the plane, no two of which are parallel, meet in a common point.” Here is a “proof” that $P(n)$ is true for all positive integers $n \geq 2$.

- BASIS STEP: The statement $P(2)$ is true because any two lines in the plane that are not parallel meet in a common point.
- INDUCTIVE STEP: The inductive hypothesis is the statement that $P(k)$ is true for the positive integer $k \geq 2$, i.e., every set of k lines in the plane, no two of which are parallel, meet in a common point.
- We must show that if $P(k)$ holds, then $P(k + 1)$ holds,
- i.e., if every set of k lines in the plane, no two of which are parallel, $k \geq 2$, meet in a common point,
- then every set of $k + 1$ lines in the plane, no two of which are parallel, meet in a common point.

continued →

An Incorrect “Proof” by Mathematical Induction (Cont.)

Inductive Hypothesis: Every set of k lines in the plane, where $k \geq 2$, no two of which are parallel, meet in a common point.

- Consider a set of $k + 1$ distinct lines in the plane, no two parallel. By the inductive hypothesis, the first k of these lines must meet in a common point p_1 . By the inductive hypothesis, the last k of these lines meet in a common point p_2 .
- If p_1 and p_2 are different points, all lines containing both of them must be the same line since two points determine a line. This contradicts the assumption that the lines are distinct. Hence, $p_1 = p_2$ lies on all $k + 1$ distinct lines, and therefore $P(k + 1)$ holds. Assuming that $k \geq 2$, distinct lines meet in a common point, then every $k + 1$ lines meet in a common point.
- There must be an error in this proof since the conclusion is absurd. But where is the error?
 - **Answer :** $P(k) \rightarrow P(k + 1)$ only holds for $k \geq 3$. It is not the case that $P(2)$ implies $P(3)$. The first two lines must meet in a common point p_1 and the second two must meet in a common point p_2 . They do not have to be the same point since only the second line is common to both sets of lines.

Template for Proofs by Mathematical Induction

1. Express the statement that is to be proved in the form “for all $n \geq b$, $P(n)$ ” for a fixed integer b . For statements of the form “ $P(n)$ for all positive integers n ,” let $b = 1$, and for statements of the form “ $P(n)$ for all nonnegative integers n ,” let $b = 0$. For some statements of the form $P(n)$, such as inequalities, you may need to determine the appropriate value of b by checking the truth values of $P(n)$ for small values of n , as is done in Example 6.
2. Write out the words “Basis Step.” Then show that $P(b)$ is true, taking care that the correct value of b is used. This completes the first part of the proof.
3. Write out the words “Inductive Step” and state, and clearly identify, the inductive hypothesis, in the form “Assume that $P(k)$ is true for an arbitrary fixed integer $k \geq b$.”
4. State what needs to be proved under the assumption that the inductive hypothesis is true. That is, write out what $P(k + 1)$ says.
5. Prove the statement $P(k + 1)$ making use of the assumption $P(k)$. (Generally, this is the most difficult part of a mathematical induction proof. Decide on the most promising proof strategy and look ahead to see how to use the induction hypothesis to build your proof of the inductive step. Also, be sure that your proof is valid for all integers k with $k \geq b$, taking care that the proof works for small values of k , including $k = b$.)
6. Clearly identify the conclusion of the inductive step, such as by saying “This completes the inductive step.”
7. After completing the basis step and the inductive step, state the conclusion, namely, “By mathematical induction, $P(n)$ is true for all integers n with $n \geq b$ ”.